

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 13 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1, b;
clrscr();
b=a++ + a++ + a++;
printf("a=%d, b=%d\n", a,b);
a=1;
b=++a + ++a + ++a;
printf("a=%d, b=%d\n", a,b);
getch();
}

/* a=4, b=3
   a=4, b=12_
*/
```

a=1

b=a++ + a++ + a++;

priority: +, =, a++

1. b = a + a + a ==> 1 + 1 + 1

2. b = 3

3. a++ ==> a=2, a++ ==> a=3, a++ ==> a=4

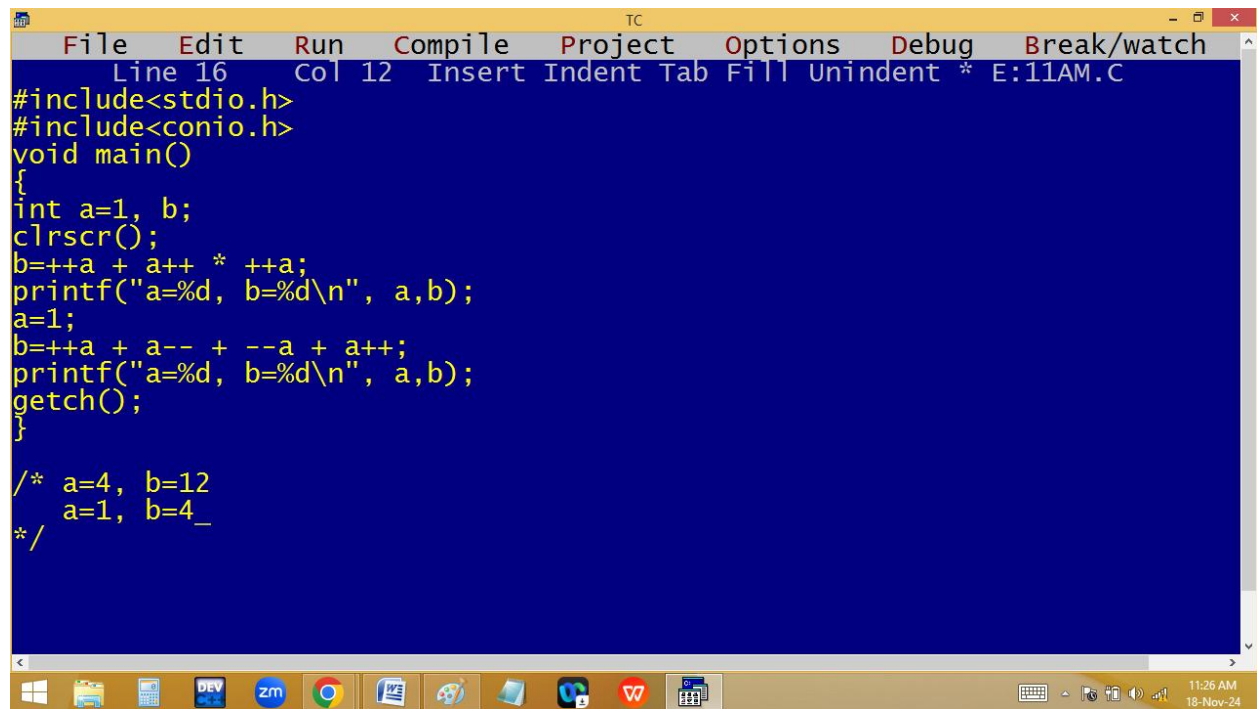
a=1

b=++a + ++a + ++a;

priority: ++a, +, =

1. ++a ==> a=2, ++a ==> a=3, ++a ==> a=4

2. b = a + a + a ==> 4 + 4 + 4 ==> b=12



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 12 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1, b;
clrscr();
b=++a + a++ * ++a;
printf("a=%d, b=%d\n", a,b);
a=1;
b=++a + a-- + --a + a++;
printf("a=%d, b=%d\n", a,b);
getch();
}

/* a=4, b=12
   a=1, b=4_
*/
```

a=1

b = ++a + a++ * ++a;

priority: ++a, *, +, =, a++

1. ++a==>a=2, ++a==>a=3

2. b = a + **a*a** ==> 3 + 3*3 ==> 3 + 9

3. **b = 12**

4. a++ ==> **a=4**

a=1

b = ++a + a-- + --a + a++;

priority: ++a, --a, +, =, a++, a--

1. ++a ==> a=2

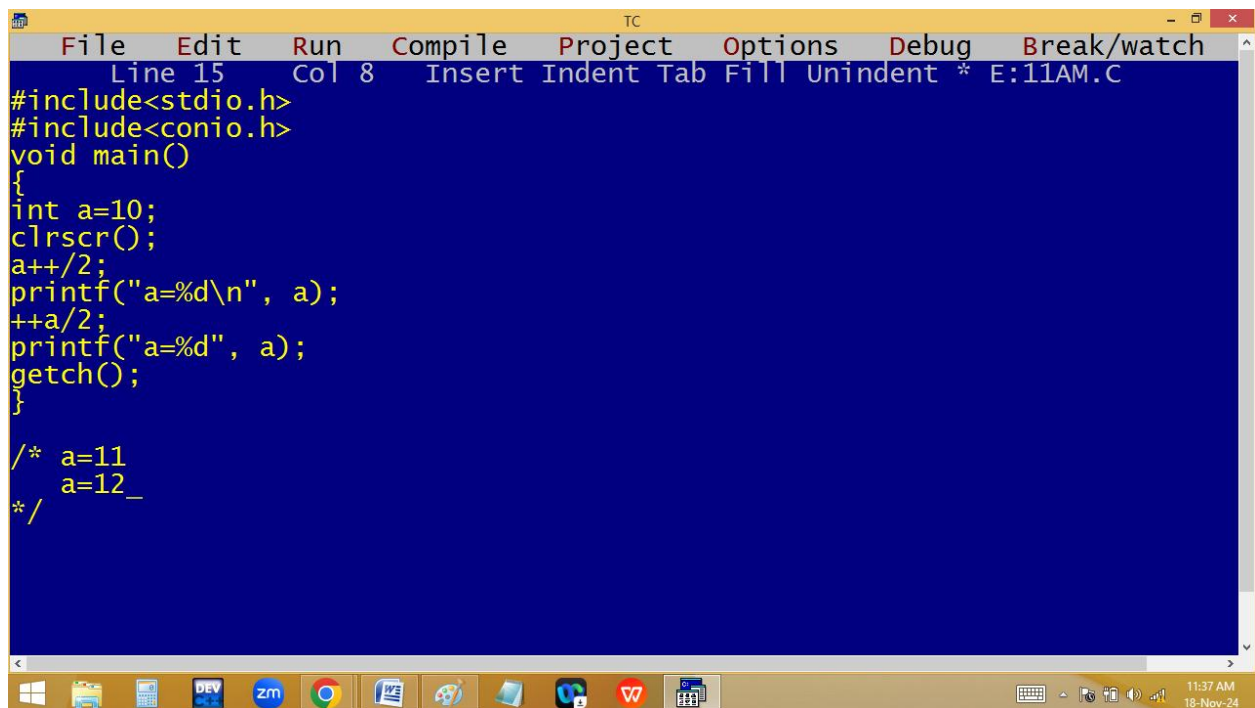
2. --a ==> a=1

3. b = a + a + a + a ==> 1 + 1 + 1 + 1

4. **b = 4**

5. a++ ==> a=2

6. a-- ==> **a=1**



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 8 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a++/2;
printf("a=%d\n", a);
++a/2;
printf("a=%d", a);
getch();
}

/* a=11
   a=12_
*/
```

a=10

a++/2;

priority: /, a++

1. a/2==>10/2=5 [5 not stored in a because of = not used i.e. a still 10]

2. a++ ==> a=11

printf(a) ==> 11

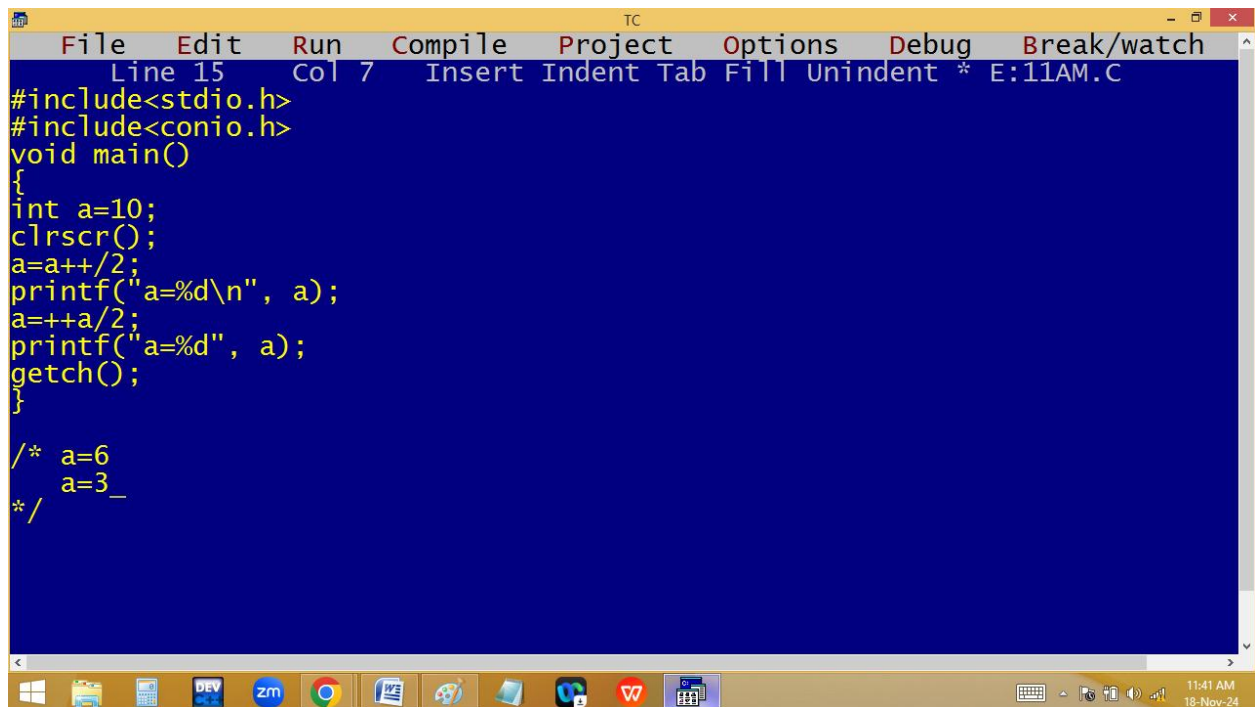
++a/2;

priority: ++a, /

1. ++a ==> a=12

2. a/2==>12/2=6 [6 not stored in a because of = not used i.e a still 12

printf(a) ==> 12



```
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 7 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a=a++/2;
printf("a=%d\\n", a);
a=++a/2;
printf("a=%d", a);
getch();
}

/* a=6
   a=3_
*/
```

a=10

a=a++/2

priority: /, =, a++

1. a=a/2 ==> 10/2 =5

2. a=5

3. a++ ==> a=6

printf(a) ==> 6

a=++a/2;

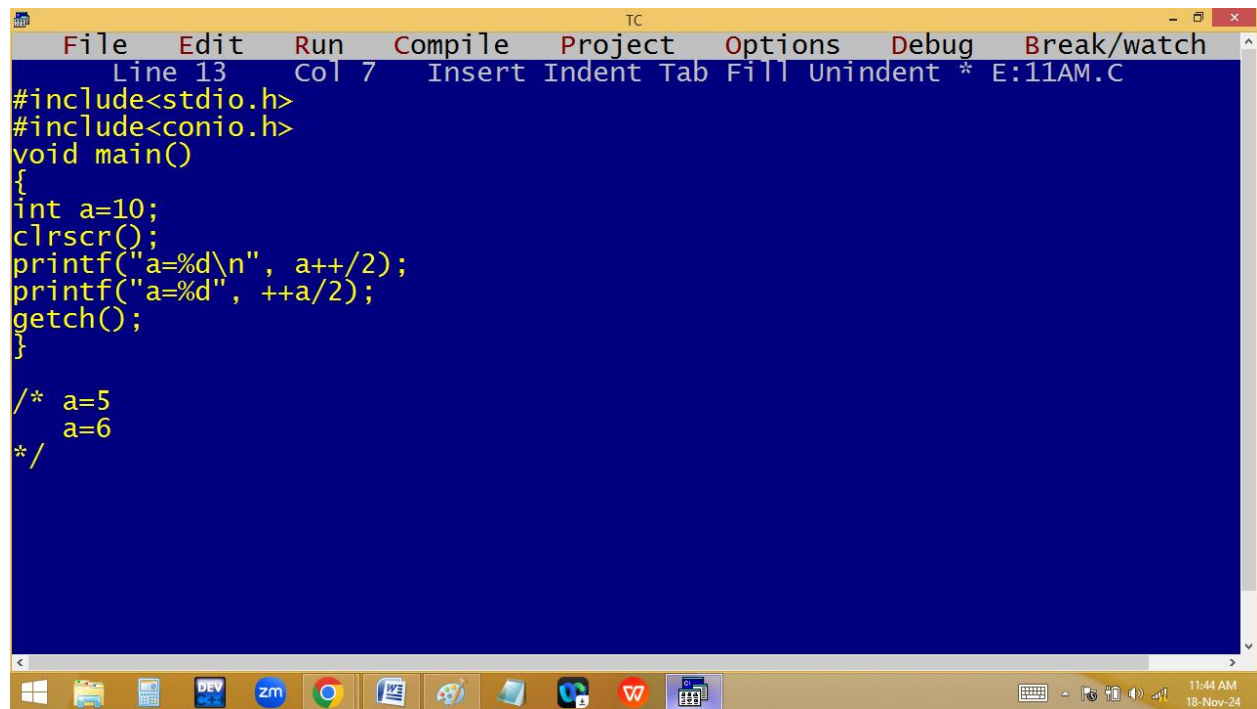
priority: ++a, /, =

1. ++a ==> a=7

2. a=a/2 ==> 7/2=3

3. a=3

printf(a) ==> a=3



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 7 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("a=%d\n", a++/2);
printf("a=%d", ++a/2);
getch();
}

/* a=5
a=6
*/
```

Windows taskbar: 11:44 AM, 18-Nov-24

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 8 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("a=%d\n", a++/2);
printf("a=%d\n", ++a/2);
printf("a=%d",a);
getch();
}

/* a=5
   a=6
   a=12_
*/
```

The bottom window shows the output of the program:

```
a=5
a=6
a=12
```

The IDE interface includes a menu bar at the top with options: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. A status bar at the bottom right indicates the time as 11:48 AM on 18-Nov-24. The taskbar at the very bottom shows various application icons including Windows Explorer, DEV, zm, Chrome, Word, and others.

a=10

printf("a=%d\n", a++/2);

priority: /, printf, a++

1. a/2==>10/2=5

2. printf(5); [5 printed not stored i.e. a=10]

3. a++ ==> a=11

printf("a=%d", ++a/2);

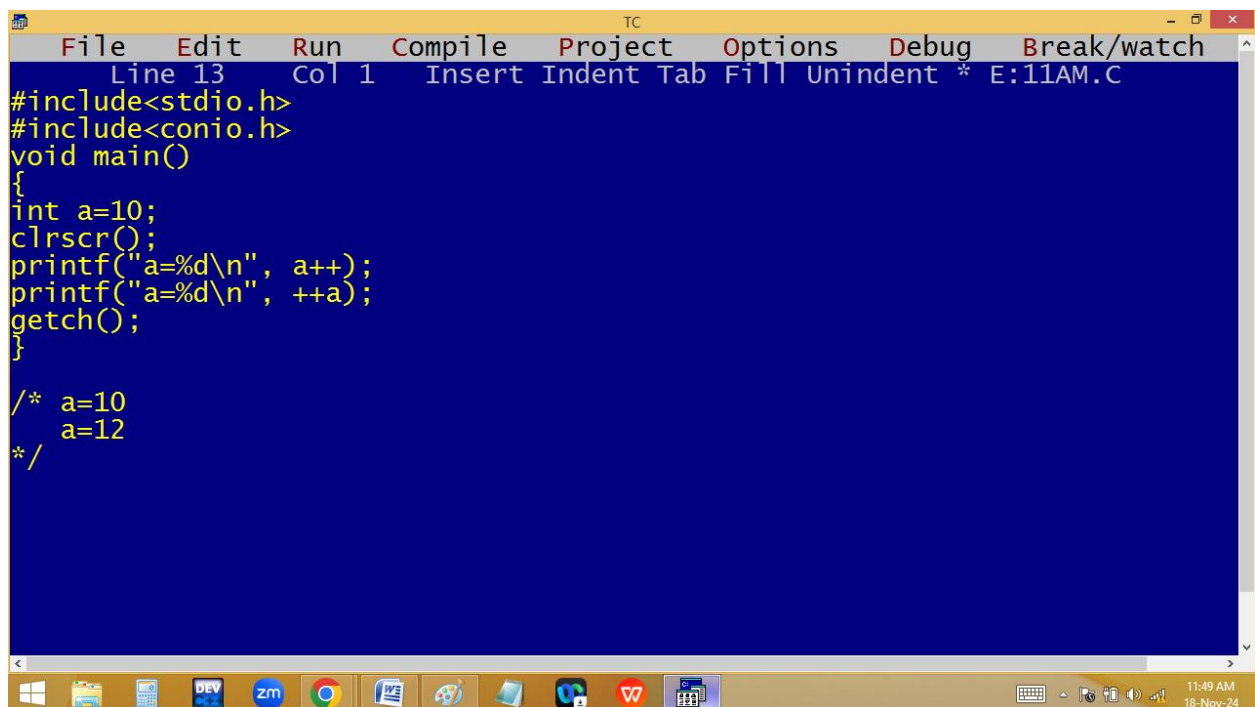
priority: ++a,/ , printf

1. ++a ==> a=12

2. a/2 ==> 12/2=6

3. printf(6) [6 printed not stored i.e. a=12]

printf(a) ==> a=12

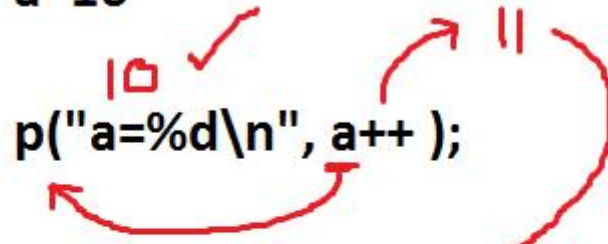


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("a=%d\n", a++);
printf("a=%d\n", ++a);
getch();
}

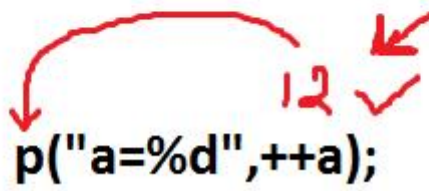
/* a=10
a=12
*/
```


a=10

p("a=%d\n", a++);



p("a=%d",++a);



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 17 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1,b=2,c;
clrscr();
c=a+++b;
printf("a=%d, b=%d, c=%d\n", a,b,c);
getch();
}

/* a=2, b=2, c=3 */
```

a=1, b=2

c = a++ + b;

1 + 2 = 3

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 12 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1,b=2,c;
clrscr();
c=a+++b++;
printf("a=%d, b=%d, c=%d\n", a,b,c);
getch();
}

/* a=2, b=3, c=3 */
```

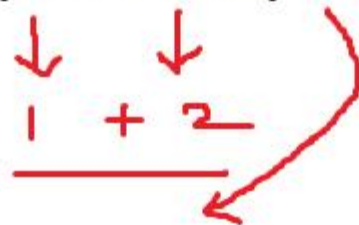
```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1,b=2,c;
clrscr();
c=(a++ + b++)++;
printf("a=%d, b=%d, c=%d\n", a,b,c);
getch();
}

/* Error */
```

a=1, b=2

c = (a++ + b++) ++;

↓ ↓
1 + 2
—



3 ++

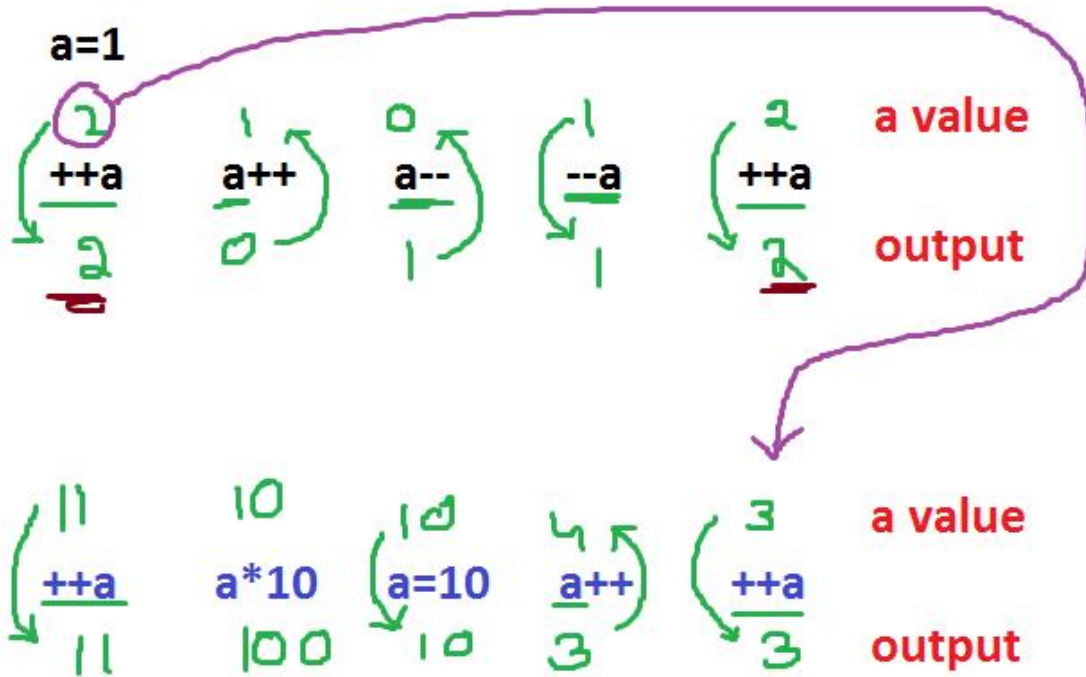
↑

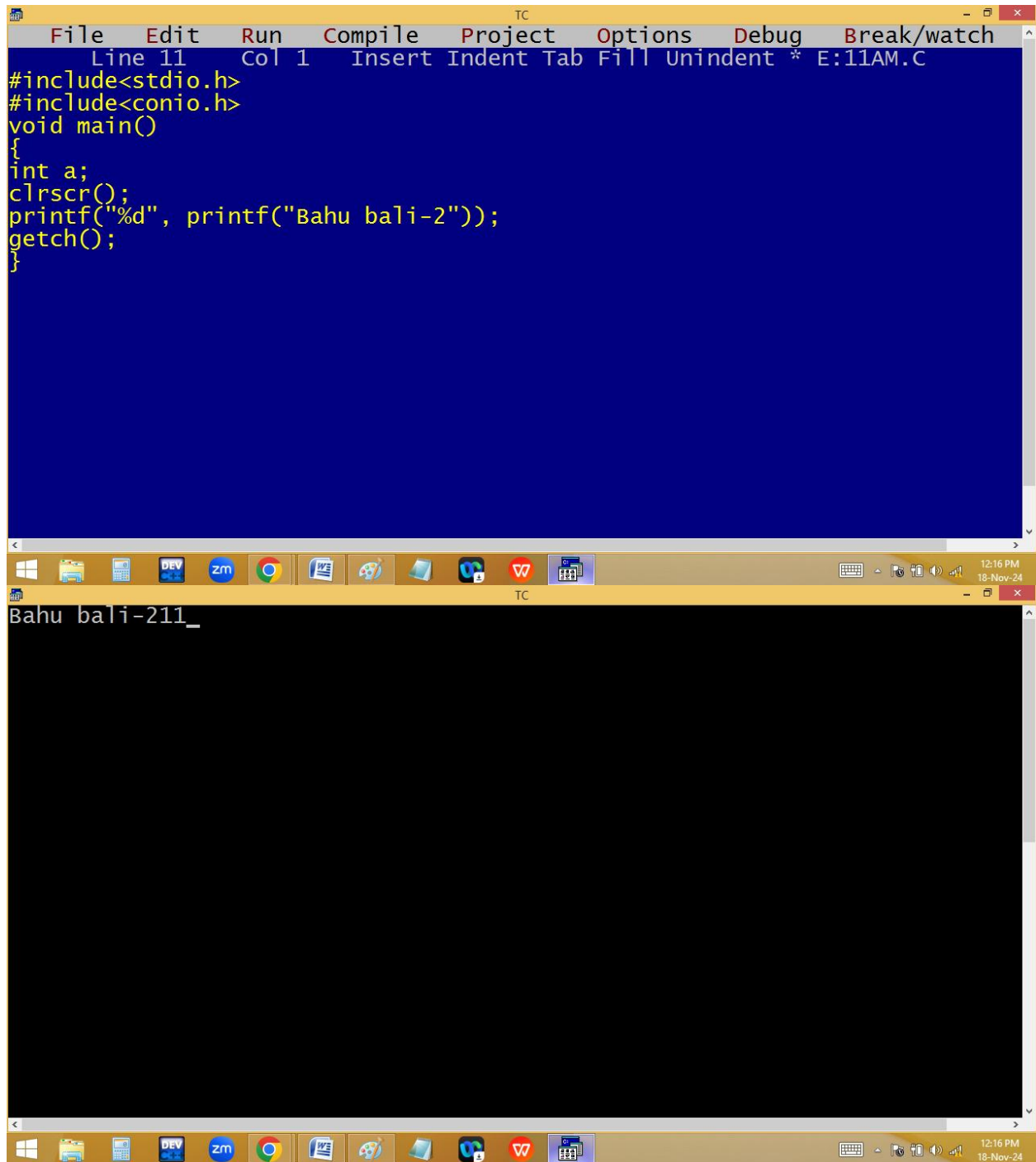
constant not changed

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=1;
clrscr();
printf("%d,%d,%d,%d,%d\n",++a,a++,a--,--a,++a);
printf("%d,%d,%d,%d,%d\n",++a,a*10,a=10,a++,++a);
getch();
}

/* In printf execution order right to left and printing left to right */

TC
2,0,1,1,2
11,100,10,3,3
```





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the source code editor, which has a blue background. It contains the following C code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
printf("%d", printf("Bahu bali-2"));
getch();
}
```

The bottom window is the output console, which has a black background. It displays the output of the program:

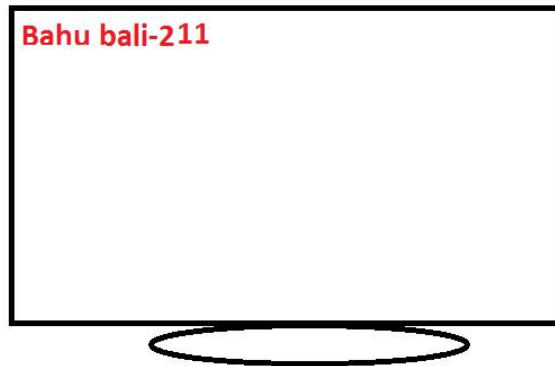
```
Bahu bali-211_
```

The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 12:16 PM on 18-Nov-24.


`printf("%d", printf("Bahu bali-2"));`
↓ ↓ ↓
2 1 Bahu bali-2



Bahu bali-211



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the editor, displaying a C program. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom of the editor shows 'Line 8', 'Col 12', and 'Insert Indent Tab Fill Unindent * E:11AM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9;
clrscr();
printf("%d\n", ++a, a=25);
printf("%d, %d, %d\n", a++, ++a, a=printf("Thank you\n"));
printf("%d",a);
getch();
}
```

The bottom window is the output console, which shows the execution results:

```
26
Thank you
11, 11, 10
12_
```

The Windows taskbar at the bottom of the screen shows the time as 12:29 PM on 18-Nov-24. Various application icons are visible on the taskbar, including DEV, zm, Chrome, and others.

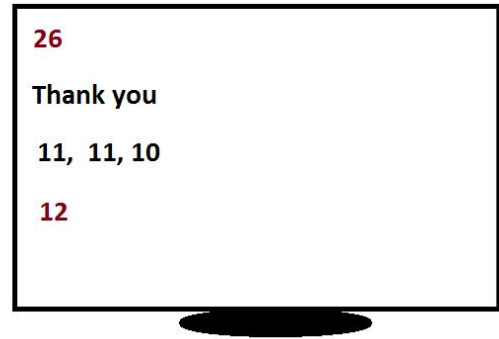
```

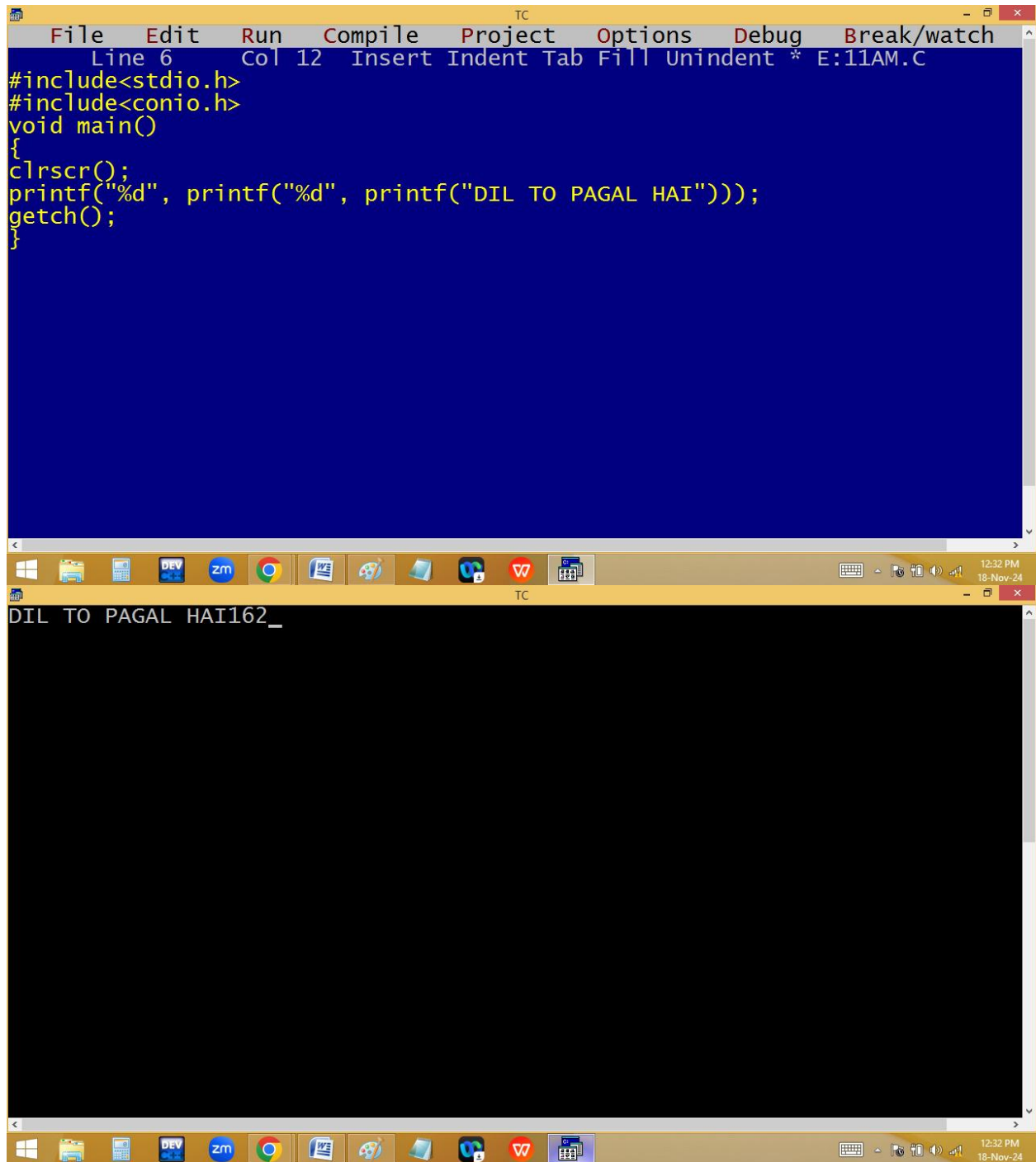
a=9;
printf("%d\n", ++a, a=25);
printf("%d, %d, %d\n", a++, ++a, a=printf("Thank you\n"));

```

Handwritten annotations:

- Red arrow from `a=9;` to `26` with a checkmark.
- Red arrow from `printf("%d\n", ++a, a=25);` to `11` with a checkmark.
- Red arrow from `printf("%d, %d, %d\n", a++, ++a, a=printf("Thank you\n"));` to `12` with a checkmark.
- Black arrow from `a++` to `11` with a checkmark.
- Black arrow from `++a` to `11` with a checkmark.
- Black arrow from `a=printf("Thank you\n")` to `10` with a checkmark.





The image shows two windows from the Turbo C++ (TC) IDE. The top window is the editor, titled 'E:11AM.C', with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 6, Col 12, Insert Indent Tab Fill Unindent *). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d", printf("%d", printf("DIL TO PAGAL HAI")));
getch();
}
```

The bottom window is the output console, titled 'TC', which displays the output of the program: 'DIL TO PAGAL HAI162_'. The Windows taskbar at the bottom shows the time as 12:32 PM on 18-Nov-24.

$\frac{2}{16}$
p("%d", p("%d", p("DIL TO PAGAL HAI"))));
16



The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'INC.CPP', contains the following C++ code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int x=3, y=2;
int z; // declaration
z=x++ * (x+y) - y--; // initialization
clrscr();
printf("x=%d, y=%d, z=%d",x,y,z);
getch();
}
```

The bottom window shows the output of the program: `x=4, y=1, z=13_`. The Windows taskbar at the bottom indicates the time is 12:39 PM on 18-Nov-24.

$x=3$ $y=2$

$z = x++ * (x+y) - y--;$

priority: (), *, -, = , x++, y--

1. $(x+y) ==> 3+2 ==> 5$

2. $z = 3 * 5 - 2$

3. $z = 15 - 2$

4. $z = 13$ ✓

5. $x++ ==> x=4$ ✓

6. $y-- ==> y=1$ ✓

The image shows a screenshot of the Turbo C++ IDE. The top window, titled 'INC.CPP', contains the following C++ code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int x=3, y=2;
int z=x++ * (x+y) - y--; // declaration & initialization
clrscr();
printf("x=%d, y=%d, z=%d",x,y,z);
getch();
}
```

The bottom window shows the output of the program: `x=4, y=1, z=16_`. The IDE's menu bar includes File, Edit, Search, Run, Compile, Debug, Project, Options, and Window. The Windows taskbar at the bottom shows the time as 12:43 PM on 18-Nov-24.

x=3 y=2

int z = x++ * (x + y) - y--;

$3 \times (4 + 2) - 2$

$3 \times 6 - 2 = 16$

Note: When declaration and initialization completed in same line, there is no priority of operators.